Towards Temporal Knowledge Graph Alignment in the Wild
Technical Report

***Abstract*—This technical report contains the dataset details and the full experimental setup of the paper "Towards Temporal Knowledge Graph Alignment in the Wild".**

I. Supplementary Information on Methodology

Algorithm 1: Overall Framework of HyDRA

Input : Source TKG G^s, Target TKG G^t
Output: Final alignment ϕ_f

1 Initialize final alignment set $\phi_f \leftarrow \emptyset$
2 Encode G^s and G^t into multi-granular representations
3 **while** *not converged and iterations* $< I_n$ **do**
4   Compute embeddings: $\{\boldsymbol{h}_n^{dw}\}$, $\{\boldsymbol{h}_n^{(g)}\}_{g\in\mathcal{G}}$, $\{\boldsymbol{h}_n^{\text{name}}\}$
5   Adaptive integration: generate $\boldsymbol{P}^{ada}$ and select top-k_1 pseudo pairs
6   Dual-channel projection: $\tilde{\mathbb{P}}^{e^{s,t}}_{time}, \tilde{\mathbb{P}}^{e^{s,t}}_{rel} \leftarrow$ Eq. (5)
7   Construct projection hypergraph $\mathcal{G}_p$ and memory bank $\mathcal{M}_p$
8   Retrieve top-k_2 projections: generate $\mathcal{G}_m = \left(\bigcup_{\ell=1}^{3}\mathcal{V}_\ell, \bigcup_{\ell=1}^{3}\mathcal{E}_\ell\right)$
9   **for** *each scale* $\ell \in \{1,2,3\}$ **do**
10   Intra-scale interaction: $\mathcal{G}_\ell^* \leftarrow \mathcal{I}_\ell(\mathcal{G}_\ell)$
11   Fusion reasoning: $\phi_\ell \leftarrow$ LLM-Select$(\mathcal{G}_\ell^*)$
12   Conflict detection: $\mathcal{D}, \{C(e_i^s)\} \leftarrow$ Eq. (14)
13   Resolve conflicts: $\phi_C \leftarrow$ LLM-Select$(\{C(e_i^s)\})$
14   Update $\phi_f \leftarrow \bigcap_{\ell=1}^{3}\phi_\ell \cup \phi_C$
15 **return** ϕ_f

A. The Workflow of HyDRA

The overall workflow of HyDRA is presented in Algorithm 1. It takes as input a source TKG G^s and a target TKG G^t, and outputs the final alignment ϕ_f. The algorithm initializes the alignment set as empty and encodes both TKGs into multi-granular representations (lines 1–2).

In the *encoding and integration stage* (lines 4–5), structural embeddings $\{\boldsymbol{h}_n^{dw}\}$, multi-granular temporal embeddings $\{\boldsymbol{h}_n^{(g)}\}_{g\in\mathcal{G}}$ where $\mathcal{G} = \{\text{year}, \text{month}, \text{date}\}$, and name embeddings $\{\boldsymbol{h}_n^{\text{name}}\}$ are computed (line 4). Adaptive integration generates the similarity matrix $\boldsymbol{P}^{ada}$ and selects top-k_1 pseudo-aligned pairs (line 5).

In the *projection and retrieval stage* (lines 6–8), dual-channel projections $\tilde{\mathbb{P}}^{e^{s,t}}_{time}$ and $\tilde{\mathbb{P}}^{e^{s,t}}_{rel}$ are computed via Eq. (5) (line 6). The projection hypergraph $\mathcal{G}_p$ and memory bank $\mathcal{M}_p$ are constructed (line 7). Top-k_2 projections are retrieved to generate the multi-scale hypergraph $\mathcal{G}_m = \left(\bigcup_{\ell=1}^{3}\mathcal{V}_\ell, \bigcup_{\ell=1}^{3}\mathcal{E}_\ell\right)$ (line 8).

In the *fusion stage* (lines 9–14), for each scale $\ell \in \{1,2,3\}$, intra-scale interaction refines $\mathcal{G}_\ell$ into $\mathcal{G}_\ell^*$ via LLM-guided supplementation and trimming (line 10), and fusion reasoning generates scale-specific alignments ϕ_ℓ (line 11). Conflicts $\mathcal{D}$ and conflict sets $\{C(e_i^s)\}$ are detected via Eq. (14) (line 12), and resolved through LLM selection to obtain ϕ_C (line 13). The final alignment is updated as $\phi_f = \bigcap_{\ell=1}^{3}\phi_\ell \cup \phi_C$ (line 14). After iterative refinement, the algorithm returns ϕ_f (line 15).

II. Dataset

In this section, we introduce several key metrics used to quantify the characteristics and differences between temporal knowledge graph alignment (TKGA) datasets for assessing the difficulty and realism.

Overlapping Rate (Overlapping.%). To better reflect the real-world scenario where cross-KG alignment is rarely strictly 1-to-1, we use the overlapping rate to measure the degree of shared temporal entities between two TKGs [1]. The overlapping rate is calculated as follows:

$$\textbf{Overlapping.\%}(G^s) = \frac{|\phi_{seed}|}{|E^s|} \times 100\%, \tag{1}$$

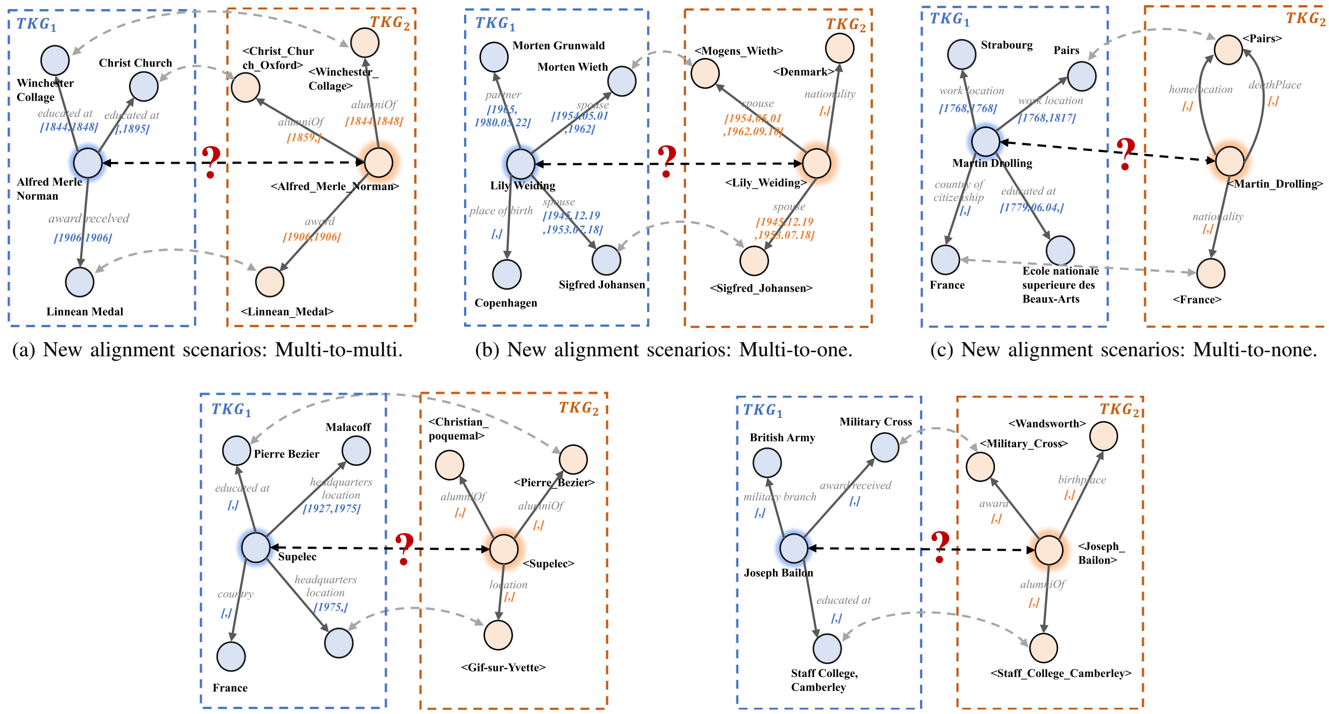
$$\textbf{Overlapping.\%}(G^t) = \frac{|\phi_{seed}|}{|E^t|} \times 100\%, \tag{2}$$

where ϕ_{seed} denotes the set of aligned entity pairs, and E^s and E^t are the set of all temporal entities in G^s and G^t.

Temporal Interval Consistency (Inter. Consis.%). To effectively assess the differences in temporal intervals between aligned entities in TKGA, we introduce a new metric called *temporal interval consistency*, which is the proportion of aligned entities with consistent temporal intervals among all aligned entities, defined as follows:

$$\begin{aligned}&\textbf{Inter.Consis.\%}(G^s, G^t) = \\ &= \frac{|(e_i^s, e_j^t) \in \phi_{seed} \mid \mathcal{T}(e_i^s) \cap \mathcal{T}(e_j^t) \neq \emptyset|}{|\phi_{seed}|} \times 100\%,\end{aligned} \tag{3}$$

where ϕ_{seed} is the set of aligned entity pairs, and $\mathcal{T}(e)$ denotes the set of time intervals associated with entity e.



(a) New alignment scenarios: Multi-to-multi. (b) New alignment scenarios: Multi-to-one. (c) New alignment scenarios: Multi-to-none.

(d) New alignment scenarios: One-to-none. (e) New alignment scenarios: None-to-none.

Fig. 1: New alignment scenarios in `BETA`.

Multi-source Valid Temporal Fact Ratio (MTF.%). To assess the extent of missing valid temporal facts in multi-source TKGs, we define the multi-source valid temporal fact ratio as follows:

$$\textbf{MTF.\%}(G^s, G^t) = \frac{|\mathcal{V}G^s + \mathcal{V}G^t|}{|G^s + G^t|} \times 100\%, \tag{4}$$

where $|\mathcal{V}G^s + \mathcal{V}G^t|$ represents the total number of valid temporal facts in the source and target TKGs, and $|G^s + G^t|$ denotes the total number of all facts in both TKGs.

Differences in Valid Temporal Facts and Valid Temporal Density (ΔT.F.% and ΔT.D.%). To better capture the imbalance of valid temporal facts and temporal structures between two TKGs in real-world TKGA scenarios, we define the relative differences in valid temporal facts and valid temporal fact density, respectively, as follows:

$$\boldsymbol{\Delta}\textbf{T.F.\%}(G^s, G^t) = \frac{|\mathcal{V}G^s - \mathcal{V}G^t|}{\min(|\mathcal{V}G^s|, |\mathcal{V}G^t|)} \times 100\%, \tag{5}$$

$$\boldsymbol{\Delta}\textbf{T.D.\%}(G^s, G^t) = \frac{|\rho^s - \rho^t|}{\min(\rho^s, \rho^t)} \times 100\%, \tag{6}$$

where $|\mathcal{V}G^s|$ and $|\mathcal{V}G^t|$ represent the number of valid temporal facts in source TKG and targe TKG, and $\rho = \frac{|\mathcal{V}G|}{|G|}$ denotes the density of valid temporal facts in a TKG.

A. New Alignment Scenarios

For existing datasets, the TKGA algorithms merely need to align the football players and teams between two KGs, as the majority of relations are *member_of_sports_team* and *plays_for*. In contrast, in `BETA` and `WildBETA`, due to the introduction of more realistic quadruples, in addition to the current alignment scenario where the entities to be aligned are only associated with one specific temporal relation, there are new temporal alignment scenarios emerging:

Multi-to-multi/multi-to-one Alignment. In this setting, at least one of the entities to be aligned is associated with multiple temporal relations. In both cases, entities are involved in multiple temporal quadruples, and it is important to mine useful signals from the complex structure while reducing the noise introduced by heterogeneous relational and temporal information.

Multi-to-none/one-to-none Alignment. In this setting, at least one of the entities to be aligned is not associated with any temporal relations. In such cases, the key is to properly address discrepancies in temporal information and sufficiently exploit features modeled from other aspects.

None-to-none alignment. In this setting, neither entity is associated with any temporal relation. As a result, temporal entity alignment is reduced to the general entity alignment task.

We present specific cases of multi-to-multi, multi-to-one, multi-to-none, one-to-none, and none-to-none alignment regarding the temporal relations in Fig. 1.

III. Supplementary Experiments

A. Experimental Setting

Datasets. In our experiments, we conducted comprehensive evaluations on **10 datasets**, including `BETA`, `WildBETA`, and

TABLE I: Dataset statistics [1]–[5]. "*#Ent*", "*#Rel.*", "*#Facts*", "*#T.Facts*": The number of entities, relations, quadruples and quadruples with valid time interval in KG1 (KG2), respectively. "*Temp.*", "*Multi-Granular*" : Indicates whether the dataset includes temporal knowledge information and the dataset includes multi-granular temporal knowledge information, respectively. "*#Overlapping*": Represents the proportion of overlapping temporal entities in KG1 and KG2. "*Inter. Consis.*" : Represents the proportion of aligned entities with consistent temporal intervals among all aligned entities. "*Δ T.F.%*", "*Δ T.D.%*" : Relative difference in valid temporal facts/density values between two KGs, using the KG with the smaller valid temporal facts/lower valid temporal density as the base. "*Multi-Source*", "*MTF.%*" : Refers to whether both TKGs in the dataset are temporal incompleteness, and the average proportion of valid temporal facts in the two TKGs, respectively.

Dataset		#Ent.	#Rel.	Temp.	Multi-Granular	#Seed	#Overlapping	Inter. Consis. ⇓	#Facts	#T.Facts	Δ T.F.% ⇑	#T.Density	Δ T.D.% ⇑	Multi-Source	MTF.% ⇓
AGROLD	AGROLD	96,117	7	✗	✗	11,555	✗	✗	49,924	✗	✗	✗	✗	✗	✗
	AGROLD	51,488	4	✗	✗		✗		364,606	✗		✗			
DOREMUS	-	2,057	19	✗	✗	238	✗	✗	6,832	✗	✗	✗	✗	✗	✗
	-	1,889	20	✗	✗		✗		5,543	✗		✗			
DBP15K(EN-FR)	EN	15,000	193	✗	✗	15,000	✗	✗	96,318	✗	✗	✗	✗	✗	✗
	FR	15,000	166	✗	✗		✗		80,112	✗		✗			
DBP-WIKI	DBP	100,000	413	✗	✗	100,000	✗	✗	293,990	✗	✗	✗	✗	✗	✗
	WIKI	100,000	261	✗	✗		✗		251,708	✗		✗			
ICEWS-WIKI	ICEWS	11,047	272	✓	✗	5,058	45.79%	55.63%	3,527,881	3,527,881	6,817.1%	319.352	9,853.4%	✗	96.05%
	WIKI	15,896	226	✓	✗		31.82%		198,257	51,002		3.208			
ICEWS-YAGO	ICEWS	26,863	272	✓	✗	18,824	70.07%	7.39%	4,192,555	4,192,555	13,764.3%	156.072	11,633.3%	✗	98.21%
	YAGO	22,734	41	✓	✗		82.80%		107,118	30,240		1.330			
DICEWS	ICEWS	9,517	247	✓	✗	8,566	90.01%	95.09%	307,552	307,552	0%	32.316	0.2%	✗	100%
	ICEWS	9,537	246	✓	✗		89.82%		307,553	307,553		32.248			
YAGO-WIKI50K	YAGO	49,629	11	✓	✗	49,172	99.08%	93.63%	221,050	221,050	43.8%	4.454	45.0%	✗	100%
	WIKI	49,222	30	✓	✗		99.90%		317,814	317,814		6.457			
BETA	WIKI	42,666	257	✓	✓	40,364	94.60%	55.12%	199,879	104,774	49.9%	2.456	48.6%	✓	48.22%
	YAGO	42,297	45	✓	✓		95.43%		162,320	69,896		1.653			
WildBETA	WIKI	27,519	301	✓	✓	17,124	62.23%	**5.27%**	527,977	142,145	**14,001.7%**	5.165	**13,722.5%**	✓	**25.37%**
	YAGO	26,975	40	✓	✓		63.48%		36,283	1,008		0.037			

8 current datasets (as detailed in Tab. I).

Among these, DICEWS and YAGO-WIKI50K are the most frequently used datasets for Temporal Knowledge Graph Alignment (TKGA), derived from ICEWS05-15, YAGO, and Wikidata. Specifically, ICEWS05-15 is constructed from the ICEWS dataset [6], which comprises political events annotated with specific dates, using a daily temporal resolution and covering the period from 2005 to 2015. Xu et al. [3] randomly partitioned the quadruples in ICEWS05-15 into two equally sized subsets, yielding the datasets DICEWS-200 (D200). The YAGO-WIKI50K datasets are similarly constructed by Xu et al. [3], who first selected the top 50,000 most frequent entities from a Wikidata subset (as extracted in [7]) and linked them to corresponding entities in YAGO. Temporal facts were then added to form two temporally enriched knowledge graphs. The resulting dataset YAGO-WIKI50K-1K contains 1000 seed entity pairs.

In contrast, ICEWS-WIKI and ICEWS-YAGO [1] represent new heterogeneous and temporal datasets, posing more realistic and challenging alignment scenarios. These datasets are characterized by significant discrepancies not only in the number of entities, relations, and triples. Furthermore, the number of seeds is not directly proportional to the total entity count, adding complexity to the temporal alignment task.

In addition, two standard non-temporal KGA datasets, DBP15K(EN-FR) and DBP-WIKI [4], are also included. DBP15K(EN-FR) focuses on cross-lingual alignment, while DBP-WIKI offers a large-scale benchmark for aligning heterogeneous KGs. Both datasets exhibit similar structural properties and high overlap (100%) in aligned entities, relations, and facts.

We also consider DOREMUS and AGROLD. DOREMUS [8] is a dataset specifically designed for the classical music domain. AGROLD [9], on the other hand, is a knowledge graph used for plant science. These datasets are included to test our model on domain-specific data outside of the general-purpose knowledge graphs mentioned above.

Baselines. Currently, no specific solutions exist for TKGA-Wild. To establish a comprehensive baseline, we introduced 29 SOTA and classic baseline methods for extensive comparison:

- MTransE [10], which introduces translation vectors to align entity embeddings across languages; and
- AlignE [11], which employs neural relation extraction to identify key relationships; and
- BootEA [11], which is one of the most competitive translation-based EA methods; and
- GCN-Align [12], which trains GCNs to embed entities of each language into a unified vector space; and
- MRAEA [13], which applies attention over local neighborhoods and relation-level meta-information; and
- RREA [14], which implements relational reflection transformations to generate relation-aware embeddings; and
- RDGCN [15], which leverages GCNs for modeling structural information within knowledge graphs.
- Dual-AMN [16], which jointly captures intra-graph and cross-graph dependencies; and
- TEA-GNN [3], which treats timestamps as link attributes, using a time-aware attention mechanism to enrich entity and relation representations; and
- TREA [17], which enhances training using neighborhood aggregation and margin-based multi-class loss; and
- STEA [18], which utilizes a temporal dictionary to guide temporal alignment; and

- Dual-Match [19], which employs a temporal encoder for unsupervised layer-wise propagation; and
- MGTEA [20], which proposes a simple yet effective multi-granular approach for temporal alignment; and
- LightTEA [21], which is a lightweight TKGA model, though its temporal component yields limited improvements on existing datasets; and
- BERT [22], utilized as a pretrained language model to initialize entity embeddings using name-based features; and
- FuAlign [23], which incorporates auxiliary information to address KG heterogeneity; and
- BERT-INT [24], which combines BERT-based augmentation with auxiliary cues for improved alignment; and
- PARIS [25], which is a probabilistic iterative method capable of aligning entities without prior alignments; and
- NMN [26], which utilizes a neighborhood matching network to capture static KG's similarities in structures; and
- SDEA [27], which is a semantic-driven embedding learning framework that addresses the semantic gap between static KGs through multiple semantic enhancement strategies; and
- Attr-Int [28], which is a simple yet effective framework designed to handle static KG by integrating attribute-level information with structural embeddings; and
- DAEA [5], which enhances static KGA by employing multi-source domain adaptation to bridge the gap between various KG distributions; and
- Simple-HHEA [1], which is a representation learning-based approach tailored for aligning heterogeneous and temporal KGs; and
- ChatEA [2], which applies large language models with fine-tuning to perform advanced KG alignment; and
- HTEA [29], which employs frequency-based temporal embeddings to enhance alignment performance; and
- Naive RAG [30], [31], a basic LLM-based RAG approach that first retrieves relevant information based on a user query and then generates answers using the retrieved content; and
- Self-Consistency [32], a chain-of-thought baseline that produces multiple reasoning paths and selects the most frequent answer as the final output. In our implementation, we further enhance it by using the top-1 most similar entity from the similarity matrix produced by Simple-HHEA as a preprocessing step for the knowledge graph; and
- Self-RAG [33], a self-reflective RAG method aimed at improving the generation quality of LLMs; and
- EasyEA † [34], an effective LLM-based method that aligns knowledge graphs through information summarization, feature fusion, and candidate selection. To ensure a uniform and equitable evaluation under identical supervision constraints, we adapt this training-free baseline into a supervised variant by learning an orthogonal linear projection matrix (i.e., Procrustes Alignment) over its fused embeddings to calibrate cross-KG space shifts before candidate selection.

Implementation details. All experiments were conducted on a server equipped with four NVIDIA GeForce RTX 4090 graphics cards, each with 24 GB of GDDR6X memory. The system features a 64-core processor and 480 GB of RAM. For storage, the server utilizes a 30 GB system disk alongside a 50 GB solid-state drive (SSD) for data storage. All implementations were carried out using the PyTorch framework.

The large language models (LLMs) reported in Tab. II and Tab. III were evaluated under identical settings, employing GPT-4[1], except for ChatEA, which directly follows the results reported in its original paper. For subsequent experiments, unless otherwise specified, GPT-3.5[2] was adopted as the default LLM owing to its cost-effectiveness.

The multi-granular information encoders and the integrated training were configured with a learning rate of 0.01, a weight decay of 0.001, gamma set to 1.0, and were trained for 500 epochs. The training set proportions follow the settings used in prior work and are set as follows: `WildBETA` (2%), `YAGO-WIKI50K-1K` (2%), `DICEWS-200` (2.3%), `BETA` (10%) [20], `ICEWS-WIKI` (30%) [1], `ICEWS-YAGO` (30%) [1], `DBP15K(EN-FR)` (30%), and `DBP-WIKI` (30%).

Evaluation metrics. Consistent with prior benchmark studies [1], [4], we adopt two widely recognized evaluation metrics to assess the effectiveness of entity alignment models: Hits@k and Mean Reciprocal Rank (MRR).

1) Hits@k evaluates the proportion of correctly aligned entity pairs that appear among the top-k ranked candidates. Formally, let N denote the total number of reference (ground truth) alignments, and for each reference entity e_i, let rank_i denote the rank position of its correct counterpart in the candidate list. The Hits@k is defined as:

$$\text{Hits@}k = \frac{1}{N}\sum_{i=1}^{N}\mathbb{I}(\text{rank}_i \leq k), \tag{7}$$

where $\mathbb{I}(\cdot)$ is the indicator function, which returns 1 if the condition is true and 0 otherwise. In practice, Hits@1 reflects the strict accuracy of top-1 predictions, while Hits@10 provides insight into broader top-k retrieval performance.

2) Mean Reciprocal Rank (MRR) measures the average of the reciprocal ranks of the correct entities in the prediction lists. It is computed as:

$$\text{MRR} = \frac{1}{N}\sum_{i=1}^{N}\frac{1}{\text{rank}_i}. \tag{8}$$

MRR captures both the presence and position of correct alignments, thereby emphasizing early correct retrieval.

Both metrics are positively oriented, meaning higher values indicate better alignment quality. Notably, in cases where models yield only the final alignment predictions (i.e., without ranked candidate lists), the Hits@1 score is substituted with standard precision.

[1] gpt-4-0125-preview from the OpenAI API, https://openai.com/api/
[2] gpt-3.5-turbo-1106 from the OpenAI API, https://openai.com/api/

TABLE II: More results on DOREMUS and AGROLD.

Methods	DOREMUS			AGROLD		
	Hits@1	Hits@10	MRR	Hits@1	Hits@10	MRR
RDGCN	0.012	0.109	-	0.001	0.003	-
NMN	0.000	0.004	-	0.001	0.001	-
SDEA	0.387	0.560	0.461	0.001	0.001	0.001
BERT-INT	0.479	0.593	0.515	0.215	0.250	0.229
Attr-Int	0.487	0.765	0.587	0.143	0.204	0.167
DAEA	<u>0.778</u>	<u>0.886</u>	<u>0.815</u>	<u>0.271</u>	<u>0.349</u>	<u>0.300</u>
HyDRA (Ours)	**0.834**	**0.903**	**0.858**	**0.432**	**0.537**	**0.455**

TABLE III: Sensitivity analysis of the retrieval size k_2 on TKGA datasets.

k_2	WildBETA	BETA	YAGO-WIKI50K-1K	DICEWS-200
3	0.801	0.856	0.954	0.975
5	0.843	0.892	0.968	0.984
8	0.866	0.916	0.975	0.988
10	**0.871**	**0.924**	**0.978**	**0.989**
12	0.869	0.923	0.977	0.988
15	0.865	0.919	0.975	0.987
20	0.861	0.915	0.973	0.986

B. Generalization on More Static KGA Scenarios

To verify the adaptability of HyDRA, we conduct additional experiments on two domain-specific datasets: DOREMUS (classical music) and AGROLD (plant science). These datasets, typically used for static or domain-adaptive alignment (e.g., DAEA [5]), lack the explicit temporal dynamics of TKGA-Wild. As shown in Tab. II, HyDRA consistently outperforms all baselines. Notably, HyDRA achieves a 59.4% relative improvement in Hits@1 on AGROLD. This superior performance demonstrates that our multi-scale hypergraph retrieval-augmented paradigm and *scale-weave synergy* mechanism are not only robust to temporal disparities but also highly effective in capturing complex structural dependencies and overcoming schema asymmetry in diverse, specialized domains.

C. Sensitivity Analysis of the Retrieval Size k_2

We conduct a sensitivity analysis about HyDRA to study the effect of the retrieval size k_2 in the relevant entity search module. Tab. III reports the Hits@1 performance under different k_2 values on four TKGA datasets.

When k_2 is small, the performance is noticeably limited, indicating that retrieving only a few projected entities provides insufficient evidence for accurate alignment. As k_2 increases from 3 to 5, Hits@1 improves substantially across all datasets, demonstrating the benefit of incorporating multiple relevant projections.

The performance becomes stable when k_2 lies in the range of 8 to 12, where only marginal fluctuations are observed. This stability is attributed to the fact that our subsequent modules perform inter-scale interaction, multi-scale fusion reasoning, and conflict detection, which collectively mitigate the sensitivity of the model to the specific choice of retrieval size. This plateau suggests that the most informative projected entities have already been retrieved and the framework effectively filters or integrates candidates.

Further increasing k_2 beyond this range does not yield additional gains and may slightly degrade performance, likely due to the introduction of less relevant or noisy projections. Based on this observation, we set $k_2 = 10$ as a representative and robust choice in all experiments.

D. Extended Analysis on Multi-Scale Hypergraph Quality and Structural Effectiveness

To thoroughly evaluate the structural utility and quality of the multi-scale hypergraph $\mathcal{G}_m$, we report its intrinsic candidate organization capabilities, its impact on downstream pseudo-labels, and the final alignment performance on the BETA and WildBETA datasets.

First, we analyze the quality of the hypergraph context prior to the downstream fusion module. Tab. IV reports the *Recall* (inclusion coverage of the ground-truth target within the candidate pool) and *Top-1 Precision* (Top-1 Prec., accuracy of the top-1 target entity before fusion module).

TABLE IV: Analysis on multi-scale hypergraph quality.

Method	BETA		WildBETA	
	Recall	Top-1 Prec.	Recall	Top-1 Prec.
HyDRA ($\mathcal{G}_m$)	**97.8%**	**79.6%**	**96.4%**	**74.2%**
- *w/o* Multi-Scale Hypergraph	88.4%	56.8%	82.5%	53.1%

As observed in Tab. IV, the full configuration HyDRA($\mathcal{G}_m$) achieves a Recall of 97.8% on BETA and 96.4% on WildBETA, with a Top-1 Precision of 79.6% and 74.2%, respectively. In contrast, removing the multi-scale hypergraph ($-$*w/o* Multi-Scale Hypergraph) turns the context into an simple candidate list, which induces a noticeable performance drop (e.g., Recall drops by 13.9% and Top-1 Precision drops by 21.1% on the asymmetrical WildBETA benchmark). This comparative gap indicates that the scale-adaptive projection and multi-scale hypergraph retrieval modules effectively filter cross-scale relational-temporal noise, ensuring the correct targets are highly concentrated.

Second, the high-quality context organized by the hypergraph directly supports the iterative updates of the downstream reasoning module. As tracked in Tab. VI (detailed in Sec. III-E), the pseudo-labels generated by the multi-scale interaction-augmented fusion module maintain a high precision of over 94% across sequential rounds. This stable precision demonstrates that the higher-order relational groupings within $\mathcal{G}_m$ provide effective structural constraints, preventing the downstream module from generating low-confidence misalignments during iterations. More detailed analysis regarding pseudo-label evolution and error propagation is provided in Sec. III-E.

Third, we verify how this multi-scale hypergraph quality impacts the final alignment metrics. Tab. V presents a systematic ablation study on the topological designs of the hypergraph.

When the multi-scale hypergraph structure is completely omitted and processed as a flat reference setting ($-$*w/o* Multi-Scale Hypergraph), the final Hits@1 accuracy decreases significantly from 0.859 to 0.791 on WildBETA and from 0.921 to 0.835 on BETA. Furthermore, restricting the topology to

TABLE V: Ablation study on hypergraph structural designs and components on WildBETA and BETA.

Settings	WildBETA			BETA		
	Hits@1	Hits@10	MRR	Hits@1	Hits@10	MRR
HyDRA	**0.859**	**0.904**	**0.878**	**0.921**	**0.944**	**0.927**
- *w/o* Multi-Scale Hypergraph	0.791	0.859	0.838	0.835	0.881	0.862
- *w/* Single-Scale Hypergraph ($\ell = 1$)	0.813	0.868	0.846	0.863	0.907	0.887

a single scale (−*w/* Single-Scale Hypergraph ($\ell = 1$)) yields moderate improvements over the flat list baseline but still consistently underperforms the complete multi-scale framework. These experimental results validate that the scale-adaptive projection and multi-scale hypergraph retrieval modules better capture complex *temporal span* and balancing *heterogeneous temporal structures*.

E. Impact of Pseudo-Label Noise and Error Propagation

Iterative TKGA inherently suffers from error propagation, where mismatched pseudo-labels can amplify across alignment rounds and degrade final precision. HyDRA is specifically designed to suppress this dynamic. In this section, we track the evolutionary quality of the pseudo-label pool and conduct a rigorous stress test via explicit noise injection on WildBETA to validate the robustness of our framework.

Pseudo-Label Quality Tracking over Iterations. Tab. VI shows the precision and downstream performance across three sequential iteration rounds. We observe that although the pseudo-label Precision experiences a marginal decay (from 96.5% to 94.8%) due to the inevitable nature of iterative error accumulation, maintaining a high precision level above 94%. Concurrently, the accumulated Hits@1 improves monotonically and substantially from 0.812 to 0.859. This trade-off demonstrates that the positive knowledge gained from incremental alignments heavily outweighs the minor noise propagation, successfully validating the efficacy of our cross-scale conflict detection gate in intercepting low-confidence candidates at early stages.

TABLE VI: Quality of generated pseudo-labels across iterations on WildBETA.

Metric	Round 1 (I_1)	Round 2 (I_2)	Round 3 (I_3)
Precision	96.5%	95.2%	94.8%
Accum. Hits@1	0.812	0.845	**0.859**
Accum. Hits@10	0.868	0.891	**0.904**
Accum. MRR	0.832	0.861	**0.878**

Robustness to Explicit Noise Injection. To further evaluate the framework's resilience under adversarial conditions, we inject controlled proportions of incorrect pairs (10%, 20%, and 30% of the pool size) into the pseudo-label pool generated in Round 1, prior to the start of Round 2. As illustrated in Tab. VII, HyDRA exhibits strong robustness, with Hits@1 degrading gracefully from 0.859 to 0.785 even under a severe 30% noise contamination. Rather than triggering a catastrophic error cascade, the model absorbs the perturbations smoothly. This resilience implies that the multi-scale hypergraph retrieval, coupled with bidirectional consistency verification, provides sufficient structural coverage and high-precision feedback to neutralize the propagation of explicitly injected localized noise.

TABLE VII: Performance of HyDRA under varying levels of injected pseudo-label noise on WildBETA.

Noise Ratio	Hits@1	Hits@10	MRR
0% (Standard HyDRA)	0.859	0.904	0.878
10% Injected Noise	0.846	0.895	0.865
20% Injected Noise	0.828	0.881	0.849
30% Injected Noise	0.785	0.852	0.806

F. Extended Analysis on LLM-based Knowledge Graph Alignment Methods

To assess the necessity of the full hypergraph RAG architecture relative to simpler LLM paradigms, we conduct a detailed comparison against ChatEA and EasyEA † on WildBETA and BETA. Tab. VIII reports both accuracy and computational cost.

Accuracy. LLM-based re-ranking (ChatEA, EasyEA †) improves over non-retrieval LLM methods (e.g., Self-Consistency) and benefits from the richer contextual structure of BETA versus WildBETA, consistent with the stronger temporal asymmetry in the latter. Notably, HyDRA consistently achieves the highest performance across both datasets (0.859 vs. 0.765 Hits@1 on WildBETA; 0.921 vs. 0.806 on BETA). This significant improvement highlights the advantage of our multi-scale hypergraph RAG architecture, which explicitly captures high-order temporal dependencies and asynchronous topological structures that standard flat re-ranking approaches typically bypass.

Efficiency. Compared to EasyEA † on WildBETA, HyDRA reduces average inference time by 86.4% (2.5 s vs. 18.4 s) and token consumption by 96.2% (318 vs. 8,293). The same trends hold on BETA. These results confirm that the proposed architecture achieves superior accuracy with substantially lower computational cost, making it practical for real-world deployment.

G. Analysis on LLM Dependency and Framework Robustness

To evaluate the architectural generalization of HyDRA, we analyze framework consistency across different LLM backbones and expanding degrees of task complexity.

Framework Robustness Across LLM Backbones. Tab. IX evaluates cross-model generalization across different LLM backbones on WildBETA and ICEWS-WIKI. The non-LLM structure-only method, Simple-HHEA, is included as an empirical reference. The experimental results demonstrate that transitioning to the smaller Llama-3-8B backbone induces performance drops across all semantic-driven frameworks. This performance drop under Llama-3-8B reflects the inherent difficulty of TKGA-Wild challenges. Under this identical open-source backbone constraint, HyDRA maintains the highest

TABLE VIII: Comprehensive accuracy and efficiency analysis of HyDRA and LLM-based baselines across WildBETA and BETA datasets.

Settings	WildBETA					BETA				
	Accuracy			Cost (Efficiency)		Accuracy			Cost (Efficiency)	
	Hits@1	Hits@10	MRR	Avg. tokens	Avg. time (s)	Hits@1	Hits@10	MRR	Avg. tokens	Avg. time (s)
Simple-HHEA *(structure)*	0.591	0.674	0.621	-	**1.3**	0.730	0.808	0.751	-	**1.2**
Naive RAG *	0.531	-	-	1,843	10.2	0.421	-	-	1,927	11.4
Self-Consistency *	0.321	-	-	13,651	20.7	0.456	-	-	14,208	22.3
Self-RAG *	0.587	-	-	34,712	30.5	0.633	-	-	35,489	31.8
ChatEA	0.712	0.818	0.745	26,384	26.8	0.755	0.852	0.789	27,156	28.4
EasyEA †	0.765	0.841	0.802	8,293	18.4	0.806	0.884	0.843	8,671	16.8
HyDRA (Ours)	**0.859**	**0.904**	**0.878**	**318**	2.5	**0.921**	**0.944**	**0.927**	**304**	2.7

TABLE IX: Cross-model generalization across different LLM backbones on WildBETA and ICEWS-WIKI.

Methods	WildBETA (Hits@1)		ICEWS-WIKI (Hits@1)	
	GPT-3.5	Llama-3-8B	GPT-3.5	Llama-3-8B
Simple-HHEA *(structure)*	0.591		0.720	
Naive RAG	0.531	0.312	0.723	0.418
Self-Consistency	0.321	0.198	0.718	0.391
Self-RAG	0.587	0.371	0.772	0.501
ChatEA	0.712	0.412	0.880	0.392
HyDRA (Ours)	**0.859**	**0.639**	**0.978**	**0.734**

alignment accuracy on both benchmarks, remaining state-of-the-art among all LLM-based methods under the same LLM backbone.

Task Complexity as the Driver of LLM Sensitivity. Tab. X monitors the performance profiles of representative frameworks under expanding degrees of task complexity. In non-TKGA-Wild scenarios such as YAGO-WIKI50K, where temporal disparities are absent, HyDRA utilizing Llama-3-8B achieves a Hits@1 of 0.971, achieving performance comparable to stronger LLM-based methods utilizing the more powerful LLM backbone.

TABLE X: Performance analysis of representative methods under expanding degrees of TKGA task complexity.

Methods	YAGO-WIKI50K		BETA		WildBETA	
	GPT-3.5	Llama-3-8B	GPT-3.5	Llama-3-8B	GPT-3.5	Llama-3-8B
Naive RAG	0.102	0.089	0.421	0.298	0.531	0.312
Self-RAG	0.175	0.158	0.633	0.441	0.587	0.371
ChatEA	0.953	0.921	0.755	0.489	0.712	0.412
HyDRA	**0.978**	**0.971**	**0.921**	**0.734**	**0.859**	**0.639**

As the dataset configurations introduce more severe TKGA challenges from YAGO-WIKI50K to BETA and WildBETA, the performance divergence between the different LLM backbones generally widens across examined methods. This performance trajectory indicates that the framework sensitivity to language model capabilities is intrinsically governed by the underlying temporal challenges of the task.

H. Robustness under Semantic Name Obfuscation

To evaluate the generalization and robustness of HyDRA when textual information is incomplete or uninformative, we conduct an evaluation on the WildBETA dataset under a name-masked setting. Specifically, we simulate a text-scarce environment by randomly masking 50% of the entity names across the knowledge graphs, replacing them with a dummy token ([MASKED]).

TABLE XI: Performance analysis under standard setting and 50% name-masked setting on WildBETA. Δ denotes the absolute performance drop compared to the standard setting.

Methods	Standard Setting			50% Name-Masked Setting		
	Hits@1	Hits@10	MRR	Hits@1 (Δ)	Hits@10	MRR
Naive RAG *	0.531	-	-	0.214 (-0.317)	-	-
Self-Consistency *	0.321	-	-	0.102 (-0.219)	-	-
Self-RAG *	0.587	-	-	0.245 (-0.342)	-	-
ChatEA	0.712	0.818	0.745	0.452 (-0.260)	0.563	0.491
HyDRA (Ours)	**0.859**	**0.904**	**0.878**	**0.815** (-0.044)	**0.875**	**0.838**

As shown in Tab. XI, the injection of a 50% name mask incurs substantial performance degradation across all LLM-based baselines. For instance, the Hits@1 of ChatEA and Self-RAG decreases by 0.260 and 0.342, respectively. This result shows that the LLM-based methods rely heavily on surface string correlations.

HyDRA demonstrates pronounced stability under identical text-scarce conditions, sustaining a marginal Hits@1 reduction of 0.044. The remaining Hits@1 of 0.815 surpasses the most competitive baseline (ChatEA) by an absolute margin of 0.363 in Hits@1 (36.3 percentage points). This resilience implies that the integration of multi-scale hypergraph modeling and temporal interval constraints yields robust structural signatures capable of compensating for textual ambiguity, mitigating the sole dependence on name semantics.

IV. Extended Evaluation on Efficiency and Scalability

A. Scalability and Feasibility for Large-Scale Temporal Knowledge Graph Alignment

Large-scale knowledge graph alignment (KGA), involving graphs with millions of entities, is widely recognized as an independent research task [35], [36], distinct from conventional KGA settings such as benchmark-scale datasets and temporal knowledge graphs including TKGA-Wild. Existing large-scale KGA systems do not attempt to perform direct alignment over the full knowledge graph due to prohibitive computational and memory requirements [35]. Instead, they follow a decomposition-based paradigm, where the original graph is partitioned into a set of smaller subgraphs, and standard KGA algorithms are applied independently within each subgraph.

This paradigm has been widely adopted in the literature. Representative examples include LIME [36], DivEA [37] and ELsEA [35], which introduce dedicated preprocessing modules such as blocking, graph partitioning, and structure-aware subgraph extraction to reduce the effective problem size. Crucially, the downstream KGA method is treated as a generic inference component, and scalability is primarily achieved through preprocessing rather than by altering the alignment algorithm itself.

Large-scale Temporal Knowledge Graph Alignment Settings. Currently, there is no dedicated research on large-scale temporal knowledge graph alignment. Most prior TKGA benchmarks contain at most tens of thousands of entities, leaving million-scale temporal alignment entirely unexplored. To fill this gap comprehensively, we conduct experiments on 10 datasets spanning three tiers. *(i) Benchmark-Scale Efficiency.* We first test on four representative datasets (ICEWS-WIKI, ICEWS-YAGO, WildBETA, and BETA) to establish efficiency baselines at conventional scale. *(ii) Million-Scale TKGA.* To evaluate HyDRA in the million-scale regime, we construct and release three large-scale TKGA datasets: TimeD1M/EN-FR (total ∼1.9M entities), TimeD1M/EN-DE (total ∼1.9M entities), and Time-FB2M (total ∼4.6M entities).*(iii) Static Large-Scale KGA Generalization.* To further verify generalization, we include three representative static large-scale KGA datasets: DBP1M/EN-FR, DBP1M/EN-DE, and FBDBP2M[3].

Following the established standard in the large-scale KGA community [35], [36], ELsEA [35] serves as the preprocessing stage to partition each large KG pair into tractable sub-tasks[4]. Moreover, on large-scale TKGA and KGA datasets, 20% of alignment seeds are used as training data, while the remaining are reserved for testing. We compare against six methods spanning non-LLM-based approaches Dual-AMN, LLM-based approaches (Naive RAG, Self-Consistency, LLM4EA, Self-RAG, ChatEA), and HyDRA. We report Hits@1/Hits@5/MRR (%), average tokens per target entity (−−− for non-LLM methods), inference time (hours), and peak local memory (GB).

Hypergraph Size and Complexity Analysis. To understand the scalability of the proposed framework HyDRA, we analyze the hypergraph size and computational complexity under large-scale TKGA tasks. Let $|E^s|$ denote the total number of source entities, and N denote the number of partitioned sub-tasks generated during the preprocessing stage. Through the balanced partitioning of the source graph, each self-contained sub-task contains approximately $|E^s_{\text{sub}}| \approx |E^s|/N$ source entities.

The hypergraph construction and subsequent iterative updates are executed within the boundaries of each individual sub-task. For a single sub-task, the scale-adaptive entity projection module operates on the localized candidate pool, generating at most $|E^s_{\text{sub}}| \times k_1 \times 2$ multidimensional projections. Consequently, the space complexity for storing the sparse hyperedge adjacency matrix per sub-task is bounded by $\mathcal{O}(|E^s_{\text{sub}}| \times k_1)$. From a computational perspective, since the alignment processes are handled within independent sub-tasks, the hypergraph retrieval and fusion reasoning can be executed in a distributed or parallel manner. This training structure restricts the maximum retrieval search space to the local sub-task capacity rather than the global entity space. Furthermore, during the iterative update rounds, verified alignment pairs are progressively removed from the candidate pool, allowing the effective size of the local hypergraphs to decrease monotonically and optimizing the computational efficiency in later iterations. The specific scalability and effectiveness can be found in the following experiments.

[3]More details can be found in https://github.com/eduzrh/HyDRA.

[4]The preprocessing pipeline is publicly integrated as a plug-in for community use: https://github.com/eduzrh/HyDRA.

Benchmark-Scale Efficiency. Tab. XII evaluates the performance and computational costs of HyDRA as the dataset scale increases from 27K to 85K entities. As observed, HyDRA obtains the highest Hits@1 scores across all four benchmarks, with the performance margin being more noticeable in complex settings. For instance, on WildBETA, HyDRA outperforms ChatEA by 0.147 and Self-RAG by 0.272 in Hits@1, whereas this gap narrows on the structurally simpler ICEWS-WIKI dataset, indicating that the multi-scale design is more helpful for TKGA-Wild scenarios. While lightweight retrieval methods (Naive RAG, Self-Consistency) generally require lower execution times than larger LLM frameworks like Self-RAG, their alignment accuracy drops on more challenging datasets, underperforming HyDRA on WildBETA by 0.328 and 0.538, respectively. Notably, HyDRA consistently maintains the lowest time costs among all evaluated methods while displaying stable memory usage, suggesting its practical adaptability within benchmark-scale environments.

TABLE XII: Benchmark-scale efficiency across datasets of increasing total entity scale.

Dataset	Scale	Method	Hits@1	Avg. time (s)	Mem
ICEWS-WIKI	27K	Naive RAG *	0.723	8.4	2.4
		Self-Consistency *	0.718	17.3	2.2
		Self-RAG *	0.772	25.3	2.6
		ChatEA	<u>0.880</u>	<u>23.7</u>	<u>2.8</u>
		HyDRA	**0.978**	**2.1**	**2.5**
ICEWS-YAGO	50K	Naive RAG *	0.712	9.1	2.6
		Self-Consistency *	0.704	19.2	2.4
		Self-RAG *	0.762	27.8	2.7
		ChatEA	<u>0.875</u>	<u>18.9</u>	<u>2.9</u>
		HyDRA	**0.972**	**2.4**	**2.7**
WildBETA	54K	Naive RAG *	0.531	10.2	2.8
		Self-Consistency *	0.321	20.7	2.5
		Self-RAG *	0.587	30.5	3.0
		ChatEA	<u>0.712</u>	26.8	<u>3.1</u>
		HyDRA	**0.859**	**2.5**	**2.9**
BETA	85K	Naive RAG *	0.421	11.4	2.9
		Self-Consistency *	0.456	22.3	2.7
		Self-RAG *	0.633	31.8	3.1
		ChatEA	<u>0.755</u>	28.4	<u>3.3</u>
		HyDRA	**0.921**	**2.7**	**3.1**

Results on Million-Scale TKGA Scenarios. As shown in Tab. XIII, HyDRA demonstrates favorable scalability and efficiency while achieving higher alignment accuracy compared to several recent LLM-based paradigms (e.g., Self-RAG, ChatEA). For instance, on the largest Time-FB2M dataset, HyDRA reduces the execution time by 82.7% (29.1 h

TABLE XIII: Efficiency and scalability on million-scale TKGA datasets (`TimeD1M/EN-FR`, `TimeD1M/EN-DE`, and `Time-FB2M`). Best results among LLM-based methods are highlighted in bold, while runner-up results are underlined.

Method	TimeD1M/EN-FR					TimeD1M/EN-DE					Time-FB2M				
	H@1	H@5	MRR	Time	Mem	H@1	H@5	MRR	Time	Mem	H@1	H@5	MRR	Time	Mem
Dual-AMN	0.523	0.654	0.572	18.3	9.8	0.474	0.603	0.521	16.1	9.6	0.488	0.616	0.534	24.7	11.2
Naive RAG	0.556	0.682	0.604	26.1	4.9	0.505	0.629	0.551	22.3	4.7	0.518	0.653	0.568	43.7	12.4
Self-Consistency	0.302	0.415	0.343	37.4	**4.5**	0.276	0.384	0.315	33.8	**4.6**	0.293	0.402	0.332	59.2	**11.9**
LLM4EA	0.594	0.718	0.639	44.3	5.2	0.543	0.667	0.589	41.6	4.8	0.562	0.691	0.609	71.8	12.5
Self-RAG	0.621	0.749	0.671	58.7	5.4	0.572	0.698	0.618	54.2	5.1	0.591	0.724	0.641	91.3	12.7
ChatEA	0.684	0.806	0.729	103.6	5.6	0.638	0.761	0.684	91.4	5.3	0.657	0.789	0.706	168.3	13.1
HyDRA	**0.818**	**0.895**	**0.846**	**20.4**	5.3	**0.775**	**0.862**	**0.808**	**17.3**	5.0	**0.793**	**0.881**	**0.829**	**29.1**	12.6

TABLE XIV: Results on static large-scale KGA datasets (`DBP1M/EN-FR`, `DBP1M/EN-DE`, and `FBDBP2M`). Best results among LLM-based methods are highlighted in bold, while runner-up results are underlined.

Method	DBP1M/EN-FR					DBP1M/EN-DE					FBDBP2M				
	H@1	H@5	MRR	Time	Mem	H@1	H@5	MRR	Time	Mem	H@1	H@5	MRR	Time	Mem
Dual-AMN	0.552	0.684	0.601	16.4	9.5	0.504	0.631	0.549	14.2	9.3	0.519	0.648	0.566	20.8	10.6
Naive RAG	0.584	0.711	0.636	23.8	4.7	0.529	0.656	0.576	21.4	4.5	0.546	0.683	0.596	38.1	11.6
Self-Consistency	0.336	0.447	0.374	33.2	**4.3**	0.307	0.413	0.341	29.7	**4.4**	0.323	0.434	0.362	51.4	**11.4**
LLM4EA	0.623	0.746	0.667	43.6	4.9	0.572	0.691	0.613	38.2	4.7	0.593	0.716	0.636	64.7	12.1
Self-RAG	0.651	0.783	0.704	57.3	5.1	0.603	0.727	0.651	51.8	4.8	0.624	0.753	0.674	81.6	12.0
ChatEA	0.716	0.841	0.763	91.7	5.3	0.669	0.793	0.716	82.6	5.0	0.694	0.823	0.741	148.6	12.4
HyDRA	**0.849**	**0.923**	**0.877**	**17.1**	5.0	**0.804**	**0.889**	**0.837**	**14.6**	4.9	**0.832**	**0.911**	**0.861**	**25.8**	12.6

vs. 168.3 h) compared to ChatEA, while achieving a 20.7% relative improvement in Hits@1. Although the multi-scale hypergraph construction introduces a slightly higher memory overhead than the lightweight retrieval baselines, our framework maintains a lower time cost than other LLM-based competitors. These empirical results indicate the effectiveness of our approach in mitigating performance bottlenecks in large-scale temporal alignment.

B. Extended Results on Static Large-Scale KGA Scenarios

To evaluate the generalization capability of the proposed framework, we also conduct experiments on three widely used static large-scale KGA benchmarks: `DBP1M/EN-FR`, `DBP1M/EN-DE`, and `FBDBP2M`. The experimental results regarding alignment accuracy, inference time, and peak memory consumption are reported in Tab. XIV. The results demonstrate that HyDRA maintains a performance advantage even in non-temporal alignment scenarios, achieving the top results in Hits@1, Hits@5, and MRR across all datasets. Specifically, it yields up to a 20.2% relative improvement in Hits@1 over the strongest LLM baseline, while achieving a lower inference time and comparable memory overhead. This performance across both static and temporal scenarios indicates the scalability and feasibility of HyDRA for real-world, million-scale alignment tasks.

V. Further Discussion and Broader Impact

A. Typical Case Analysis and Future Directions

We analyze several typical misalignment cases from `WildBETA` to reveal remaining challenges and future improvement directions under TKGA-Wild settings.

Typical Case 1. Consider the entity pair (*Rodrigo Vergilio* in KG_1, *Careca* in KG_2), referring to the same Brazilian footballer under his legal name and nickname respectively. The quadruples in KG_1 carry year-month timestamps: `(Rodrigo Vergilio, member_of_sports_team, Ventforet Kofu, 2004-01)`, `(..., Cerezo Osaka, 2008-01)`, `(..., Qadsia SC, 2011-01)`, and `(..., Al Nasr SC, 2011-01)`. The corresponding quadruples in KG_2 record only year-level or undated entries for the same clubs: `(Careca, memberOf, Ventforet Kofu, 2004)`, `(..., Cerezo Osaka, ~)`, `(..., Qadsia SC, ~)`. The temporal neighbourhoods of the two entities differ in both density and granularity, and the incomplete temporal annotations on the KG_2 side further weaken the structural signal available for matching. Existing methods rely on symmetric temporal neighbourhood correspondence to compute entity similarity, cause the two entity representations to diverge substantially, yielding a low similarity score and a misalignment. HyDRA addresses this by jointly modeling heterogeneous temporal neighbourhoods and incomplete temporal evidence within a unified hypergraph-based framework, thereby effectively identifying the entity correspondence and alleviating the alignment challenges posed by TKGA-Wild settings.

Typical Case 2. Consider the entity pair (*Alé Cipollini* in KG_1, *UAE Team ADQ* in KG_2), referring to the same UCI Women's professional cycling team at two different temporal snapshots: the team competed under the Alé Cipollini sponsorship during 2011–2019 and was subsequently rebranded as UAE Team ADQ following a change of title sponsor. The quadruples in KG_1 record pre-rebrand rider memberships with explicit timestamps: `(Tatiana Guderzo, memberOf, Alé Cipollini, 2011)`, `(Soraya Paladin, memberOf, Alé Cipollini, 2017)`, `(Eri Yonamine, memberOf, Alé Cipollini, 2019)`. The quadruples in KG_2 record the same riders under the new entity name, but with incomplete temporal information: `(Eri Yonamine, memberOf, UAE Team ADQ, ~)`, `(Tatiana Guderzo, memberOf, UAE Team ADQ, ~)`, `(Soraya Paladin, memberOf, UAE Team ADQ, ~)`. The quadruples in KG_1 carry explicit timestamps spanning 2011–2019, whereas all quadruples in KG_2 exhibit temporal structural incompleteness, and the two entities thus exhibit completely non-overlapping temporal footprints. This multi-to-none alignment scenario (cf. II-A New Alignment Scenarios) further widens the gap between the two sides. Existing methods rely on temporal neighbourhood correspondence to compute entity similarity, and faced with such temporal span mismatch and structural incompleteness, they fail to capture the underlying entity correspondence, yielding a low similarity score and a misalignment. This remains an open challenge for future TKGA-Wild research.

Future Improvement Directions. The TKGA-Wild setting exposes a broader class of open challenges inherent to real-world temporal knowledge graph alignment. Looking forward, a promising direction is to develop alignment models capable of reasoning over entity evolution across time, rather than treating each KG as a static snapshot of the world. In addition, grounding alignment decisions in external event knowledge (e.g., entity rename or merger records) offers a principled avenue to bridge the gap between intra-KG structural evidence and real-world entity dynamics.

B. Further Discussion on Cost and Environmental Considerations of LLM Usage

Recent advances in LLMs have significantly expanded the modeling capacity of knowledge-driven systems. However, their growing computational cost and environmental impact have also raised increasing concerns. In this section, we provide a supplementary discussion on the cost–performance trade-offs and environmental considerations of incorporating LLMs into the proposed TKGA-Wild framework.

Cost and Efficiency Considerations. Although HyDRA leverages LLMs in the intra-scale interaction and multi-scale fusion reasoning modules, it is explicitly designed to minimize unnecessary LLM invocations. Unlike conventional LLM-based RAG pipelines that repeatedly query the LLM for large candidate sets, our framework restricts LLM usage to a small number of high-confidence candidates identified through multi-scale hypergraph retrieval. This design substantially reduces both token consumption and inference latency.

As reported in Sec. 5.5 of the main paper, HyDRA achieves a reduction of up to 99% in token usage and over 85% in execution time compared to representative LLM-based RAG baselines, while simultaneously delivering superior alignment accuracy. These results indicate that the performance gains of HyDRA do not rely on excessive LLM computation, but rather on effective retrieval, structured reasoning, and multi-scale aggregation.

Environmental Impact and Sustainable Deployment. From an environmental perspective, reduced token consumption and shorter inference time directly translate into lower energy usage and carbon emissions during deployment. While accurately quantifying the carbon footprint of proprietary LLMs remains challenging due to limited disclosure of infrastructure details, token usage and runtime are widely recognized as practical and informative proxies for environmental impact.

Our empirical results suggest that HyDRA significantly mitigates the environmental burden typically associated with LLM-based systems. Moreover, experiments in Sec. 5.6 demonstrate that the proposed framework maintains strong performance even when powered by lighter LLMs, highlighting its reduced dependence on very large models and its potential for more sustainable real-world deployment.

Overall, this supplementary analysis underscores that HyDRA strikes a favorable balance between effectiveness and efficiency. By tightly integrating retrieval, hypergraph reasoning, and selective LLM utilization, the proposed framework not only advances state-of-the-art performance on TKGA-Wild, but also aligns with emerging best practices for responsible and sustainable use of large language models.

References

[1] X. Jiang, C. Xu, Y. Shen, Y. Wang, F. Su, Z. Shi, F. Sun, Z. Li, J. Guo, and H. Shen, "Toward practical entity alignment method design: Insights from new highly heterogeneous knowledge graph datasets," in *Proceedings of the ACM on Web Conference 2024, WWW 2024, Singapore, May 13-17, 2024*, T. Chua, C. Ngo, R. Kumar, H. W. Lauw, and R. K. Lee, Eds. ACM, 2024, pp. 2325–2336.

[2] X. Jiang, Y. Shen, Z. Shi, C. Xu, W. Li, Z. Li, J. Guo, H. Shen, and Y. Wang, "Unlocking the power of large language models for entity alignment," in *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), ACL 2024, Bangkok, Thailand, August 11-16, 2024*, L. Ku, A. Martins, and V. Srikumar, Eds. Association for Computational Linguistics, 2024, pp. 7566–7583.

[3] C. Xu, F. Su, and J. Lehmann, "Time-aware graph neural network for entity alignment between temporal knowledge graphs," in *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing, EMNLP 2021, Virtual Event / Punta Cana, Dominican Republic, 7-11 November, 2021*. Association for Computational Linguistics, 2021, pp. 8999–9010.

[4] Z. Sun, Q. Zhang, W. Hu, C. Wang, M. Chen, F. Akrami, and C. Li, "A benchmarking study of embedding-based entity alignment for knowledge graphs," *Proc. VLDB Endow.*, vol. 13, no. 11, pp. 2326–2340, 2020.

[5] L. Yang, S. Zhou, J. Cheng, F. Zhang, J. Wan, S. Wang, and M. Lee, "DAEA: enhancing entity alignment in real-world knowledge graphs through multi-source domain adaptation," in *Proceedings of the 31st International Conference on Computational Linguistics, COLING 2025, Abu Dhabi, UAE, January 19-24, 2025*, O. Rambow, L. Wanner, M. Apidianaki, H. Al-Khalifa, B. D. Eugenio, and S. Schockaert, Eds. Association for Computational Linguistics, 2025, pp. 5890–5901.

[6] J. Lautenschlager, S. Shellman, and M. Ward, "ICEWS Event Aggregations," 2015.

[7] T. Lacroix, G. Obozinski, and N. Usunier, "Tensor decompositions for temporal knowledge base completion," in *8th International Conference on Learning Representations, ICLR 2020, Addis Ababa, Ethiopia, April 26-30, 2020*. OpenReview.net, 2020.

[8] P. Lisena, M. Achichi, P. Choffé, C. Cecconi, K. Todorov, B. Jacquemin, and R. Troncy, "Improving (re-)usability of musical datasets: An overview of the DOREMUS project," *CoRR*, vol. abs/2405.03382, 2024.

[9] A. Venkatesan, G. Tagny, N. El Hassouni, I. Chentli, V. Guignon, C. Jonquet, M. Ruiz, and P. Larmande, "Agronomic linked data (agrold): a knowledge-based system to enable integrative biology in agronomy," *bioRxiv*, 2018.

[10] M. Chen, Y. Tian, M. Yang, and C. Zaniolo, "Multilingual knowledge graph embeddings for cross-lingual knowledge alignment," in *IJCAI*, 2017, pp. 1511–1517.

[11] Z. Sun, W. Hu, Q. Zhang, and Y. Qu, "Bootstrapping entity alignment with knowledge graph embedding," in *Proceedings of the Twenty-Seventh International Joint Conference on Artificial Intelligence, IJCAI 2018, July 13-19, 2018, Stockholm, Sweden*. ijcai.org, 2018, pp. 4396–4402.

[12] Z. Wang, Q. Lv, X. Lan, and Y. Zhang, "Cross-lingual knowledge graph alignment via graph convolutional networks," in *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing, Brussels, Belgium, October 31 - November 4, 2018*. Association for Computational Linguistics, 2018, pp. 349–357.

[13] X. Mao, W. Wang, H. Xu, M. Lan, and Y. Wu, "MRAEA: an efficient and robust entity alignment approach for cross-lingual knowledge graph," in *WSDM '20: The Thirteenth ACM International Conference on Web Search and Data Mining, Houston, TX, USA, February 3-7, 2020*. ACM, 2020, pp. 420–428.

[14] X. Mao, W. Wang, H. Xu, Y. Wu, and M. Lan, "Relational reflection entity alignment," in *CIKM '20: The 29th ACM International Conference on Information and Knowledge Management, Virtual Event, Ireland, October 19-23, 2020*. ACM, 2020, pp. 1095–1104.

[15] Z. Chen, Y. Wu, Y. Feng, and D. Zhao, "Integrating manifold knowledge for global entity linking with heterogeneous graphs," *Data Intelligence*, vol. 4, no. 1, pp. 20–40, 2022.

[16] X. Mao, W. Wang, Y. Wu, and M. Lan, "Boosting the speed of entity alignment 10 ×: Dual attention matching network with normalized hard sample mining," in *WWW '21: The Web Conference 2021, Virtual Event / Ljubljana, Slovenia, April 19-23, 2021*. ACM / IW3C2, 2021, pp. 821–832.

[17] C. Xu, F. Su, B. Xiong, and J. Lehmann, "Time-aware entity alignment using temporal relational attention," in *WWW '22: The ACM Web Conference 2022, Virtual Event, Lyon, France, April 25 - 29, 2022*. ACM, 2022, pp. 788–797.

[18] L. Cai, X. Mao, M. Ma, H. Yuan, J. Zhu, and M. Lan, "A simple temporal information matching mechanism for entity alignment between temporal knowledge graphs," in *Proceedings of the 29th International Conference on Computational Linguistics, COLING 2022, Gyeongju, Republic of Korea, October 12-17, 2022*. International Committee on Computational Linguistics, 2022, pp. 2075–2086.

[19] X. Liu, J. Wu, T. Li, L. Chen, and Y. Gao, "Unsupervised entity alignment for temporal knowledge graphs," in *Proceedings of the ACM Web Conference 2023, WWW 2023, Austin, TX, USA, 30 April 2023 - 4 May 2023*, Y. Ding, J. Tang, J. F. Sequeda, L. Aroyo, C. Castillo, and G. Houben, Eds. ACM, 2023, pp. 2528–2538.

[20] W. Zeng, J. Zhou, and X. Zhao, "Benchmarking challenges for temporal knowledge graph alignment," *Proceedings of the 33rd ACM International Conference on Information and Knowledge Management*, 2024.

[21] L. Cai, X. Mao, Y. Xiao, C. Wu, and M. Lan, "An effective and efficient time-aware entity alignment framework via two-aspect three-view label propagation," in *Proceedings of the Thirty-Second International Joint Conference on Artificial Intelligence, IJCAI 2023, 19th-25th August 2023, Macao, SAR, China*. ijcai.org, 2023, pp. 5021–5029.

[22] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," in *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*. Minneapolis, Minnesota: Association for Computational Linguistics, 2019, pp. 4171–4186.

[23] C. Wang, Z. Huang, Y. Wan, J. Wei, J. Zhao, and P. Wang, "FuAlign: Cross-lingual entity alignment via multi-view representation learning of fused knowledge graphs," *Inform. Fusion*, vol. 89, pp. 41–52, Jan. 2023.

[24] X. Tang, J. Zhang, B. Chen, Y. Yang, H. Chen, and C. Li, "BERT-INT: A bert-based interaction model for knowledge graph alignment," in *Proceedings of the Twenty-Ninth International Joint Conference on Artificial Intelligence, IJCAI 2020*, C. Bessiere, Ed. ijcai.org, 2020, pp. 3174–3180.

[25] F. M. Suchanek, S. Abiteboul, and P. Senellart, "Paris: Probabilistic alignment of relations, instances, and schema," *Proceedings of the VLDB Endowment*, vol. 5, no. 3, 2011.

[26] Y. Wu, X. Liu, Y. Feng, Z. Wang, and D. Zhao, "Neighborhood matching network for entity alignment," in *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics, ACL 2020, Online, July 5-10, 2020*, D. Jurafsky, J. Chai, N. Schluter, and J. R. Tetreault, Eds. Association for Computational Linguistics, 2020, pp. 6477–6487.

[27] Z. Zhong, M. Zhang, J. Fan, and C. Dou, "Semantics driven embedding learning for effective entity alignment," in *38th IEEE International Conference on Data Engineering, ICDE 2022, Kuala Lumpur, Malaysia, May 9-12, 2022*. IEEE, 2022, pp. 2127–2140.

[28] L. Yang, J. Cheng, C. Xu, X. Wang, J. Li, and F. Zhang, "Attr-int: A simple and effective entity alignment framework for heterogeneous knowledge graphs," in *IEEE International Conference on Acoustics, Speech and Signal Processing, ICASSP 2024, Seoul, Republic of Korea, April 14-19, 2024*. IEEE, 2024, pp. 6315–6319.

[29] J. Li, W. Hua, F. Jin, and X. Li, "HTEA: heterogeneity-aware embedding learning for temporal entity alignment," in *Proceedings of the Eighteenth ACM International Conference on Web Search and Data Mining, WSDM 2025, Hannover, Germany, March 10-14, 2025*, W. Nejdl, S. Auer, M. Cha, M. Moens, and M. Najork, Eds. ACM, 2025, pp. 982–990.

[30] Q. Zhang, S. Chen, Y. Bei, Z. Yuan, H. Zhou, Z. Hong, J. Dong, H. Chen, Y. Chang, and X. Huang, "A survey of graph retrieval-augmented generation for customized large language models," *CoRR*, vol. abs/2501.13958, 2025.

[31] X. Ma, Y. Gong, P. He, H. Zhao, and N. Duan, "Query rewriting in retrieval-augmented large language models," in *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing, EMNLP 2023, Singapore, December 6-10, 2023*, H. Bouamor, J. Pino, and K. Bali, Eds. Association for Computational Linguistics, 2023, pp. 5303–5315.

[32] X. Wang, J. Wei, D. Schuurmans, Q. V. Le, E. H. Chi, S. Narang, A. Chowdhery, and D. Zhou, "Self-consistency improves chain of thought reasoning in language models," in *The Eleventh International Conference on Learning Representations, ICLR 2023, Kigali, Rwanda, May 1-5, 2023*. OpenReview.net, 2023.

[33] A. Asai, Z. Wu, Y. Wang, A. Sil, and H. Hajishirzi, "Self-rag: Learning to retrieve, generate, and critique through self-reflection," in *The Twelfth International Conference on Learning Representations, ICLR 2024, Vienna, Austria, May 7-11, 2024*. OpenReview.net, 2024.

[34] J. Cheng, C. Lu, L. Yang, G. Chen, and F. Zhang, "Easyea: Large language model is all you need in entity alignment between knowledge graphs," in *Findings of the Association for Computational Linguistics, ACL 2025, Vienna, Austria, July 27 - August 1, 2025*, W. Che, J. Nabende, E. Shutova, and M. T. Pilehvar, Eds. Association for Computational Linguistics, 2025, pp. 20 981–20 995.

[35] Q. Zhou, W. Chen, L. Zhang, P. Zhao, J. Xu, and L. Zhao, "Enhancing large-scale entity alignment with critical structure and high-quality context," in *41st IEEE International Conference on Data Engineering, ICDE 2025, Hong Kong, May 19-23, 2025*. IEEE, 2025, pp. 2227–2239.

[36] W. Zeng, X. Zhao, X. Li, J. Tang, and W. Wang, "On entity alignment at scale," *VLDB J.*, vol. 31, no. 5, pp. 1009–1033, 2022.

[37] B. Liu, W. Hua, G. Zuccon, G. Zhao, and X. Zhang, "High-quality task division for large-scale entity alignment," in *Proceedings of the 31st ACM International Conference on Information & Knowledge Management, Atlanta, GA, USA, October 17-21, 2022*, M. A. Hasan and L. Xiong, Eds. ACM, 2022, pp. 1258–1268. [Online]. Available: https://doi.org/10.1145/3511808.3557352